

Penrith Selective High School

March 2015

Higher School Certificate

Half Yearly Examination.

Extension 1 Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- A table of standard integrals is provided towards the back of this paper
- In Questions 6-9, show relevant mathematical reasoning and/or calculations
- All diagrams are not to scale

Total Marks – 65

Section I Pages 2–3

5 marks

- Attempt Questions 1–5
- Allow about 8 minutes for this section
- Answer these questions on the multiple choice answer sheet provided at the back of this paper

Section II Pages 4–7

60 marks

- Attempt Questions 6–9
- Allow about 2 hours 52 minutes for this section
- Start each question on a new page

Teacher Use Only	Questions 1-5	Question 6	Question 7	Question 8	Question 9	Total	%
Marks	/5	/15	/15	/15	/15	/65	

Student Name: _____

Section I:

5 marks

Attempt Questions 1–5

Allow about 8 minutes for this section

Use the multiple-choice answer sheet for Questions 1–5.

1. Simplify $\frac{\log_b(9a^2)}{\log_b(3a)}$, where $a > 0, b > 0$
- (A) 2
- (B) $3a$
- (C) $\log_b(3a)$
- (D) $\log_b(9a^2 - 3a)$
2. If α, β and γ are the roots of $x^3 - 4x^2 + 2x + 3 = 0$, what is the value of:
 $\alpha\beta + \beta\gamma + \gamma\alpha$?
- (A) -2
- (B) -4
- (C) 4
- (D) 2
3. If $P(x)$ has degree $n + 1$, then $P(x) = 0$:
- (A) always has n turning points
- (B) has at most n turning points
- (C) always has $n + 1$ turning points
- (D) has at most $n + 1$ turning points

Section 1 continued on next page.

4. Which of the following is an expression for $\int \cos^2 8x \, dx$?
- (A) $\frac{x}{2} - \frac{1}{32} \sin 8x + c$
- (B) $\frac{x}{2} + \frac{1}{32} \sin 8x + c$
- (C) $\frac{x}{2} - \frac{1}{32} \sin 16x + c$
- (D) $\frac{x}{2} + \frac{1}{32} \sin 16x + c$
5. How many numbers greater than 3000 can be formed with the digits 2, 3, 4, 5 and 6 if no digit is used more than once in the number? (You do not have to use all of the digits).
- (A) 96
- (B) 120
- (C) 196
- (D) 216

END OF SECTION I
Section 2 begins on next page.

Section II

60 Marks

Attempt Questions 6–9

Allow about 1 hour and 52 minutes for this section

Start each question on a new page. Extra paper is available.

In Questions 6–9, your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (15 marks) Start a new page.

Marks

- a) Find the acute angle between the two lines whose equations are $9x - 2y = 8$ and $11x + 7y - 12 = 0$. (3)
- b) i) Expand $(a - x)^3$ (2)
- ii) Use your expansion to show that $(1.98)^3 = 7.762$ when rounded off to 3 decimal places. (2)
- c) The polynomial $(x - a)^3 + b$ is zero at $x = 1$ and, when divided by x , the remainder is -7 . Find all possible values of a and b . (4)
- d) M and N have co-ordinates $(-1, 7)$ and $(5, -2)$ respectively. P divides MN in the ratio $k:1$.
- i) Find an expression for the co-ordinate of P . (2)
- ii) Hence, find the value of k when P lies on the line $5x - 4y - 1 = 0$. (2)

End of Question 6
Question 7 on next page

Question 7 (15 marks) Start a new page.

Marks

a) If $y = \ln(\cos x)$, show that: $\frac{dy}{dx} = -\tan x$ (2)

b) i) Sketch $y = |2x - 1|$ and $y = |2x + 1|$ on the same set of axes, showing key features. (3)

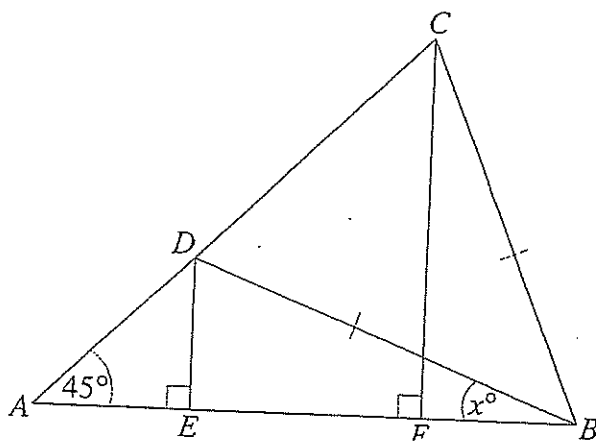
ii) Hence, find all real numbers x , such that $|2x - 1| < |2x + 1|$ (1)

c) $P(x_1, y_1)$ is any point on the curve $y = x^n$. PM and PN are perpendiculars to the x and y -axes respectively, with M and N lying on the x and y -axes respectively.

i) Draw a diagram, marking all the information given above. (1)

ii) Show that the area enclosed between the curve, the x -axis and PM is $\frac{1}{n+1}$ of the area of the rectangle $OMPN$. (O is the origin). (3)

d) $\angle BAC = 45^\circ$, $DE \perp AB$, $CF \perp AB$, $BC = BD$, $\angle ABD = x^\circ$



Prove that:

i) $\triangle BDE \equiv \triangle CBF$ (3)

ii) $AE = FB$ (2)

End of Question 7
Question 8 on next page.

Question 8 (15 marks). Start a new page.

Marks

- a) Evaluate $\tan \left[\cos^{-1} \left(\frac{3}{5} \right) \right]$ (2)
- b) i) Show that $\frac{x^2+8}{x^2+4}$ can be written as $1 + \frac{4}{x^2+4}$ (1)
- ii) Hence evaluate $\int_0^2 \frac{x^2+8}{x^2+4} dx$ (2)
- c) i) For what values of x is the function $\sin^{-1}x$ defined? (1)
- ii) Find the maximum value of $2x(1-x)$ (1)
- iii) Find the range of the function $f(x) = \sin^{-1}[2x(1-x)]$ with domain $0 \leq x \leq 1$ (2)
- d) The stationary points on the graph of $y = \frac{12x-x^3}{8}$ are given to be (2,2) and (-2,-2). (This can be assumed).
- i) A function is defined by $f(x) = \frac{12x-x^3}{8}$, for $-2 \leq f(x) \leq 2$. (3)
Sketch a graph of $y = f(x)$ and explain why an inverse function $f^{-1}(x)$ exists.
- ii) Sketch $y = f^{-1}(x)$ on the same set of axes. (1)
- iii) Find the value of $\int_0^2 f^{-1}(x) dx$ (2)

End of Question 8
Question 9 on next page

Question 9 (15 marks) Start a new page.

Marks

- a) If $t = \tan \frac{x}{2}$, express as simply as possible in terms of t , $\frac{1+\cos x}{1-\cos x}$ (2)
- b) i) Show that $8\cos x - 15\sin x$ can be written as $17\cos(x + 61^\circ 56')$. (3)
(Where $61^\circ 56'$ has been rounded off correct to the nearest minute).
- ii) Hence, find the maximum value of $8\cos x - 15\sin x$, and the smallest positive value of x for which this maximum occurs. Give your answer correct to the nearest minute. (2)
- c) i) Show that $1 + \sin 2A$ can be written as $(\cos A + \sin A)^2$ (2)
- ii) Prove that $\sqrt{\frac{1-\sin 2x}{1+\sin 2x}} = \frac{1-\tan x}{1+\tan x}$ (3)
- iii) Hence, show that the exact value of $\tan \frac{\pi}{8} = \sqrt{2} - 1$ (3)

End of Paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Student Name: _____

Multiple Choice Answer Sheet

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A  B  C  D 

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word 'correct' and drawing an arrow as follows.

A  B  C  D 

correct (arrow pointing to B)

Start here $\longrightarrow$

- | | | | | |
|----|-------------------------|-------------------------|-------------------------|-------------------------|
| 1. | A ☐ | B ☐ | C ☐ | D ☐ |
| 2. | A ☐ | B ☐ | C ☐ | D ☐ |
| 3. | A ☐ | B ☐ | C ☐ | D ☐ |
| 4. | A ☐ | B ☐ | C ☐ | D ☐ |
| 5. | A ☐ | B ☐ | C ☐ | D ☐ |

Q6. a) $9x - 2y = 8$

$$y = \frac{9x - 8}{2}$$

$$m_1 = \frac{9}{2}$$

$$11x + 7y - 12 = 0$$

$$y = \frac{-11x + 12}{7}$$

$$m_2 = -\frac{11}{7}$$

$$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$$

(Learn this formula)

$$\tan \theta = \left| \frac{\frac{9}{2} - -\frac{11}{7}}{1 - \frac{9}{2} \times \frac{11}{7}} \right|$$

$$= |1|$$

$$\theta = \tan^{-1}(1) = \underline{\underline{45^\circ \text{ or } \frac{\pi}{4}}}$$

b) i) $(a - x)^3 = \underline{a^3 - 3a^2x + 3ax^2 - x^3}$

(ii) $(1.98)^3 = (2 - 0.02)^3$

$$a = 2, x = 0.02$$

$$(1.98)^3 = 2^3 - 3 \times 2^2 \times 0.02$$

$$+ 3 \times 2 \times 0.02^2 - 0.02^3$$

$$= 8 - 0.24 + 0.0024 - 0.002^3$$

$$= 7.762392 \quad (\text{Put this line in})$$

$$= \underline{\underline{7.762}} \quad (\text{to 3 dp})$$

1A 2D 3B

4D 5D



$$6 \text{ c) } p(1) = 0$$

$$(1-a)^3 + b = 0$$

$$1 - 3a + 3a^2 - a^3 + b = 0 \quad (1)$$

$$p(0) = -7 \text{ (Remainder Theorem)}$$

Many students forgot the remainder theorem

$$(1-a)^3 + b = -7 \quad (2)$$

Sub (2) into (1)

$$1 - 3a + 3a^2 - 7 = 0$$

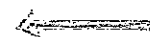
$$3a^2 - 3a - 6 = 0$$

$$a^2 - a - 2 = 0$$

$$(a-2)(a+1) = 0$$

$$a = 2, \quad b = -7 + 8 = 1$$

$$\text{or } a = -1, \quad b = -7 + -1 = -8$$



$$\begin{array}{ccc}
 & M & N \\
 i) & (-1, 7) & (5, -2) \\
 & x_1, y_1 & x_2, y_2 \\
 & K : 1 & \\
 & m : n &
 \end{array}
 \quad P(x, y)$$

$$x = \frac{n x_1 + m x_2}{m + n} = \frac{1 \times -1 + k \times 5}{k + 1}$$

$$x = \frac{5k - 1}{k + 1}$$

[Again, students were not remembering formula correctly]

$$y = \frac{1 \times 7 + k \times -2}{k + 1} = \frac{7 - 2k}{k + 1}$$

$$P\left(\frac{5k - 1}{k + 1}, \frac{7 - 2k}{k + 1}\right)$$

$$ii) 5x - 4y - 1 = 0$$

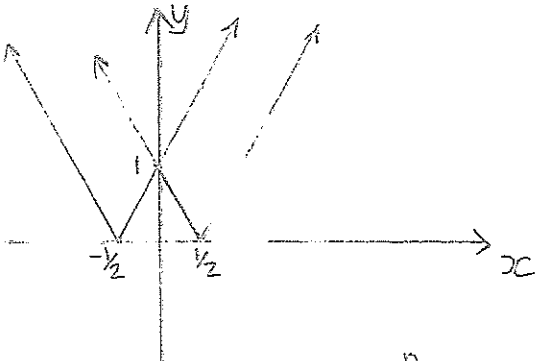
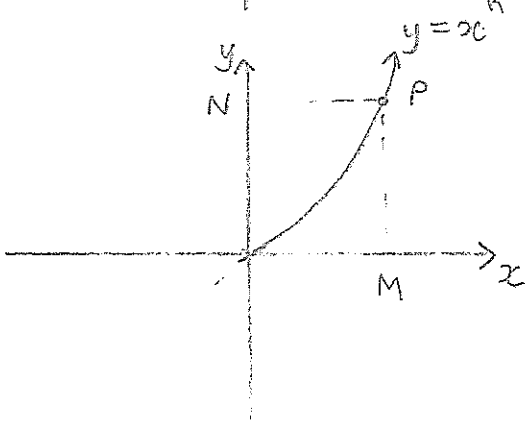
Sub P into this

$$5\left(\frac{5k - 1}{k + 1}\right) - 4\left(\frac{7 - 2k}{k + 1}\right) - 1 = 0$$

$$\frac{25k - 5 - 28 + 8k - k - 1}{k + 1} = 0$$

$$32k - 34 = 0$$

$$k = \boxed{\frac{17}{16}}$$

Exam	MATHEMATICS : Question..... Seven	
Suggested Solutions		Marker's Comments
a. $\frac{dy}{dx} = \frac{-\sin x}{\cos x}$ $= -\tan x$ bi.  ii. from graph $x > 0$ c.  Area under curve : $A = \int_0^M x^n dx$ $= \left[\frac{x^{n+1}}{n+1} \right]_0^M = \frac{M^{n+1}}{n+1}$ Area rectangle = $M \times N$ ($N = M^n$) $= M \times M^n$ $= M^{n+1}$ Area rectangle $\times \frac{1}{n+1} = \frac{M^{n+1}}{n+1}$ = Area under curve		graph goes through origin

$$\angle DCE = \angle DCF = \angle CBF$$

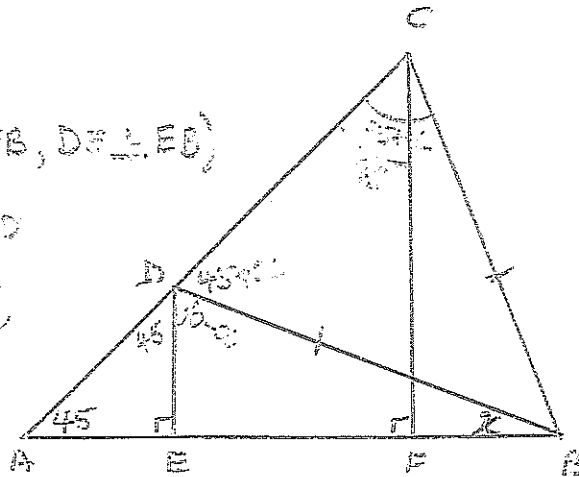
$$CB = CE \text{ (given)}$$

$$\angle CFB = \angle CED = 90^\circ \text{ (CF} \perp \text{FB, DE} \perp \text{EB)}$$

$$\angle BDE = 180^\circ - (\angle B + \angle EBD)$$

$$\angle FCA = 180^\circ - (\angle B + \angle ECA)$$

$$\begin{aligned} \angle CDB &= 180^\circ - (45^\circ + 135^\circ) \\ &= 45^\circ \end{aligned}$$



(Angles in a straight line)

$$\triangle BCD \text{ is isosceles (BD = DC)}$$

$$\angle BDC = \angle CBD = 45^\circ \text{ (equal angles opposite to equal sides)}$$

$$\angle DCF = \angle ADB = 45^\circ \text{ (corresponding angles of parallel lines)}$$

$$\therefore \angle BCF = x \text{ (sum of adjacent angles)}$$

$$\therefore \angle E = \angle BCF = x$$

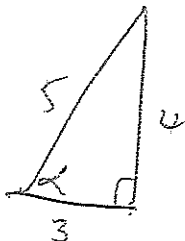
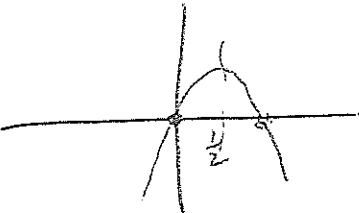
$$\therefore \triangle BDE \cong \triangle CBF \text{ (AAS)}$$

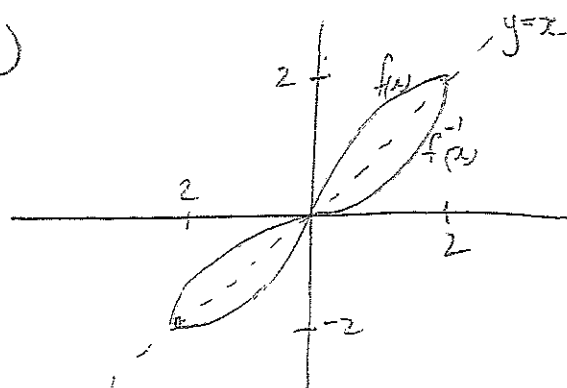
$$\text{ii } AE = DF \text{ (equal angles of corresponding triangles)}$$

$$DE = FB \text{ (corresponding sides of congruent triangles)}$$

$$\therefore AE = FB$$

Students
to write
reasons for
each step
of the
proof

Exam	Ext 1 Half yearly	MATHEMATICS	: Question.....	
Suggested Solutions				Marker's Comments
a)	$\cos^{-1} \frac{3}{5} = 2$			well done
	$\tan \alpha = \frac{4}{3}$			
b) (i)	$\frac{x^2+8}{x^2+4} = \frac{x^2+4+4}{x^2+4}$ $= \frac{x^2+4}{x^2+4} + \frac{4}{x^2+4}$ $= 1 + \frac{4}{x^2+4}$			well done
(ii)	$\int_0^2 1 + \frac{4}{x^2+4}$ $= [x]_0^2 + 2 \left[\tan^{-1} \frac{x}{2} \right]_0^2$ $= 2 + \frac{\pi}{2}$			well done except ^{some} students used wrong method to integrate $\frac{4}{x^2+4}$
c) (i)	$-1 \leq x \leq 1$			well done
(ii)	$2x(1-x)$			students who used calculus create a lot of work
	$f\left(\frac{1}{2}\right) = 1 \times \frac{1}{2}$ $= \frac{1}{2}$ $\therefore$ max value is $\frac{1}{2}$			
(iii)	$\sin^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{6}$ Range $0 \leq 2x(1-x) \leq \frac{1}{2}$ $\sin^{-1}(0) =$ $0 \leq y \leq \frac{\pi}{6}$	$\therefore$ max $\sin^{-1}[2x(1-x)]$ can reach		Students who didn't unde concept got this wrong

(d)
(i)

each y value of $f(x)$ corresponds to a unique x value.

Graphs generally needed to be larger and clearer.

$$(ii) \int_0^2 f^{-1}(x) dx \dots$$

$$\int_0^2 \frac{12x - x^3}{8} dx$$

$$\frac{1}{8} \left[6x^2 - \frac{x^4}{4} \right]_0^2$$

$$= \frac{1}{8} \left[\left(6(4) - \frac{16}{4} \right) - 0 \right]$$

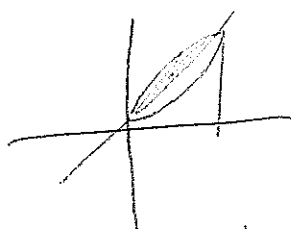
$$= \frac{1}{8} (20)$$

$$= \frac{20}{8}$$

$$\int_0^2 x dx$$

$$\left[\frac{x^2}{2} \right]_0^2$$

$$= 2$$



$$\frac{20}{8} - 2 = \frac{1}{2} \text{ (the area of shaded bit)}$$

$$\therefore 2 - \frac{1}{2} = 1\frac{1}{2} \text{ unit}^2$$

since the reflection of this area around $y=x$ is the area we need to subtract from triangle area.

Students used a variety of methods to find area. Some only got part way.

Exam	Ext 1 Half Yearly	MATHEMATICS	: Question...9
Suggested Solutions			Marker's Comments
a) $\tan \frac{x}{2}$ $\frac{1 + \frac{1-t^2}{1+t^2}}{1 - \frac{1-t^2}{1+t^2}} = \frac{1+t^2+1-t^2}{1+t^2} \div \frac{1+t^2-1+t^2}{1+t^2}$ $= \frac{2}{2t^2}$ $= \frac{1}{t^2}$			well done some algebraic errors
b(i) $8 \cos x - 15 \sin x \equiv A \cos(x + \alpha)$ $\equiv A(\cos x \cos \alpha - \sin x \sin \alpha)$ $\equiv A \cos x \cos \alpha - A \sin x \sin \alpha$ $A \cos \alpha = 8$ $A \sin \alpha = 15$ $\tan \alpha = \frac{15}{8}$ $\alpha = 61^\circ 56'$ $A^2 = 8^2 + 15^2$ $A = 17$ $\therefore 8 \cos x - 15 \sin x \equiv 17 \cos(x + 61^\circ 56')$			well done
(ii) Max value = 17 (since max value of $\cos$ is 1) this happens when $\cos(x + 61^\circ 56') = 1$ i.e. when $x + 61^\circ 56' = 0$ or 360 $\therefore x = -61^\circ 56' \text{ or } 298^\circ 4'$ $\therefore x = 298^\circ 4' \text{ (smallest positive value)}$			well done

$$C(i) \quad 1 + \sin 2A = (\cos A + \sin A)^2$$

$$\begin{aligned} \text{R.H.S.} &= \cos^2 A + 2 \sin A \cos A + \sin^2 A \\ &= 1 + 2 \sin A \cos A \\ &= 1 + \sin 2A. \end{aligned}$$

well done

$$(ii) \quad \text{L.H.S.} = \sqrt{\frac{1 - \sin 2x}{1 + \sin 2x}}$$

$$= \sqrt{\frac{(\cos x - \sin x)^2}{(\cos x + \sin x)^2}} \quad \left[\text{using part (i)} \right]$$

$$= \frac{\cos x - \sin x}{\cos x + \sin x}$$

$$= \frac{\frac{\cos x}{\cos x} - \frac{\sin x}{\cos x}}{\frac{\cos x}{\cos x} + \frac{\sin x}{\cos x}}$$

$\left[\begin{array}{l} \div \text{ top and} \\ \text{bottom by} \\ \cos x \end{array} \right]$

$$= \frac{1 - \tan x}{1 + \tan x}$$

students who failed to see how to use part (i) created a lot of work, and generally failed.

$$(iii) \quad \text{R.H.S.} = \frac{1 - \tan x}{1 + \tan x}$$

$$\tan\left(\frac{\pi}{4} - \frac{\pi}{8}\right) = \frac{\tan \frac{\pi}{4} - \tan \frac{\pi}{8}}{1 + \tan \frac{\pi}{4} \tan \frac{\pi}{8}}$$

$$\tan \frac{\pi}{8} = \frac{1 - \tan \frac{\pi}{8}}{1 + \tan \frac{\pi}{8}}$$

$$\begin{aligned} \text{L.H.S.} &= \sqrt{\frac{1 - \sin \frac{\pi}{4}}{1 + \sin \frac{\pi}{4}}} \\ \text{let } x &= \frac{\pi}{8} \end{aligned}$$

$$\begin{aligned} &= \sqrt{\frac{1 - \frac{1}{\sqrt{2}}}{1 + \frac{1}{\sqrt{2}}}} \\ &= \sqrt{\frac{\sqrt{2} - 1}{\sqrt{2} + 1}} \\ &= \sqrt{\frac{(\sqrt{2} - 1)^2}{1}} \\ &= \sqrt{2} - 1 \\ &= \tan \frac{\pi}{8}. \end{aligned}$$

students needed to show left hand side